# STAT6611

- Lecture 04 (R, Sept 3)
- Homework 1 assigned
  - practice on g.i.2 matrix results
  - due F, Sept 18 (two weeks + 1 day) by midnight in gradescope/canvas
- My office hours: TW 12:30-1:30 PM
  & R 1-2 PM

TA Ben: TBA

## 2.2 Estimability

**Definition.** An estimator $t(\underline{y})$ of a linear function $\underline{\lambda}'\underline{\beta}$ is said to be linear if $t(\underline{y}) \equiv c + \underline{a}'\underline{y}$ for some constant $c$ and some vector $\underline{a}$ of constants.

Context: you get pick $\underline{\lambda}'$ & $\underline{\beta}$ is unknown parameter (in $\mathbb{R}^p$) & $\underline{\lambda}'\underline{\beta}$ is parametric function"

**Definition.** Under Models II—VII, an estimator $t(\underline{y})$ is said to be an unbiased estimator of a linear function $\underline{\lambda}'\underline{\beta}$ if $E[t(\underline{y})] \equiv \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta} \in \mathbb{R}^p$ ← regardless of $\underline{\beta} \in \mathbb{R}^p$

Note that, under Models II—VII, $E(c + \underline{a}'\underline{y}) =$ $c + \underline{a}'E(\underline{Y}) = c + \underline{a}'X\underline{\beta}$ for any $c \in \mathbb{R}$ & $\underline{a} \in \mathbb{R}^n$

→ $\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$, $E(\underline{\varepsilon}) = 0$

**Lemma 2.1.** *Under Models II—VII, a linear estimator $c + \underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if $c = 0$ and $\underline{\lambda}' = \underline{a}'X$.*

⇐ assume $\underline{\lambda}' = \underline{a}'X$ & $c = 0$

$E(c + \underline{a}'\underline{Y}) = \underline{a}'X\underline{\beta} = \underline{\lambda}'\underline{\beta}$ for any $\underline{\beta}$ ⇒ $\underline{a}'\underline{Y}$ is unbiased

⇒ If $c + \underline{a}'\underline{y}$ is unbiased for $\underline{\lambda}'\underline{\beta}$

then $\underline{\lambda}'\underline{\beta} = E(c + \underline{a}'\underline{Y}) = c + \underline{a}'X\underline{\beta}$ for all $\underline{\beta} \in \mathbb{R}^p$

If $\underline{\beta} = 0$, then $c = 0$

If pick $\underline{\beta}$ to be columns of $I_{p\times p}$, get $\underline{a}'X I_{p\times p} = \underline{\lambda}' I_{p\times p}$

$\underline{a}'X = \underline{\lambda}'$

**Definition.** Under Models II—VII, a linear function $\underline{\lambda}'\underline{\beta}$ is said to be estimable if there exists a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Otherwise, it is said to be nonestimable.

estimable ⇔ use $\underline{a}'\underline{Y}$ where $\underline{\lambda}' = \underline{a}'X$ holds for some $\underline{a}$

⇔ $\underline{\lambda}'$ is a lin. combination of rows of $X$

**Theorem 2.1.** *Under Models II—VII, a necessary and sufficient condition for a linear function* $\underline{\lambda}'\underline{\beta}$ *to be estimable is that its coefficient vector* $\underline{\lambda}'$ *be expressible in the form* $\underline{\lambda}' = \underline{a}'X$*, that is, that* $\underline{\lambda}'$ *belong to* $\mathcal{R}(X)$*, that is, that* $\underline{\lambda}'\underline{\beta}$ *be expressible as* $\underline{a}'X\underline{\beta}$*, that is, that*

$$\operatorname{rank}(X) = \operatorname{rank}\begin{bmatrix} X \\ \underline{\lambda}' \end{bmatrix},$$

↑ row space of $X$ (i.e. $\underline{\lambda}' \in \mathcal{R}(X)$)

*where* $\begin{bmatrix} X \\ \underline{\lambda}' \end{bmatrix}$ *is the matrix* $X$ *augmented by the row vector* $\underline{\lambda}'$*.*

*Proof.* Let $\underline{u}_i'$ be the $i$th row of $X$ and $a_i$ be the $i$th element of $\underline{a}$. Then,

$$X = \begin{pmatrix} \underline{u}_1' \\ \vdots \\ \underline{u}_n' \end{pmatrix}, \ \underline{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \text{ and } \underline{a}'X = (a_1, \ldots, a_n)\begin{pmatrix} \underline{u}_1' \\ \vdots \\ \underline{u}_n' \end{pmatrix} = a_1\underline{u}_1' + \cdots + a_n\underline{u}_n' \in \mathcal{R}(X).$$

$n \times p$ $\quad$ $n \times 1$

Thus, $\underline{\lambda}' = \underline{a}'X \iff \underline{\lambda}' \in \mathcal{R}(X) \iff \operatorname{rank}(X) = \operatorname{rank}\begin{bmatrix} X \\ \underline{\lambda}' \end{bmatrix}$.

↑ ↑

This result is just restatement of Lemma 2.1

**Corollary 1.** Under Models II—VII, the elements of the vector $X\underline{\beta}$ are estimable.

$i$th row of $X$ (say $\underline{u}_i'$) $\in \mathcal{R}(X)$ so that $\underline{u}_i'\underline{\beta}$ is estimable (this is $i$th component $X\underline{\beta}$)

**Corollary 2.** Under Models II—VII, if $\underline{\lambda}_1'\underline{\beta}, \ldots, \underline{\lambda}_r'\underline{\beta}$ are estimable, then any linear combination of them is estimable.

$\underline{\lambda}_i'\underline{\beta}$ estimable $\Rightarrow \underline{\lambda}_i' = \underline{a}_i'X$ for some $\underline{a}_i$

So, if $c_1, \ldots, c_r \in \mathbb{R}$, $\sum_{i=1}^r c_i\underline{\lambda}_i'\underline{\beta} = \sum_{i=1}^r c_i\underline{a}_i'X\underline{\beta} = \underline{\lambda}'\underline{\beta}$ for $\underline{\lambda}' = \sum_{i=1}^r c_i\underline{a}_i'X = \underline{a}'X$

**Corollary 3.** If $p^* = p$, then for every $p \times 1$ vector $\underline{\lambda}$, $\underline{\lambda}'\underline{\beta}$ is estimable.

$\operatorname{rank}(X)$ → # of col of $X$ (or $\underline{\beta} \in \mathbb{R}^p$)

if $p^* = p$, then the row space of $X$ has rank $p$

every $\underline{\lambda}'$ is then $\underline{\lambda}' = \underline{a}'X$ for some $\underline{a}$

We next add a couple of technical matrix results for later use.

**Lemma 2.2.** *For any matrices $A$ and $B$ (of appropriate dimensions),* $\text{rank}(AB) \leq \text{rank}(A)$ *and* $\text{rank}(AB) \leq \text{rank}(B)$.

i.e. ranks are smaller for products compared to single matrices

Proof:

If $A=0$, result holds.

Suppose $A$ has $\text{rank}(A)=r \geq 1$.

We can diagonalize $A$ as

$$PAQ = \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix}, \text{ where } P, Q, D_r \text{ invertible}$$

($D_r$ is diagonal)

So, $\text{rank}(AB) = \text{rank}(PAB)$

($P$ invert.) $= \text{rank}(PAQQ^{-1}B)$

$$= \text{rank}\left(\begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} Q^{-1}B\right)$$

$\leq$ # of non-zero rows in $\begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix}$

$= \text{rank}(D_r)$

$= r = \text{rank}(A)$ //

**Lemma 2.3.** *For any matrix $A$,* $\text{rank}(A'A) = \text{rank}(A)$.

lemma 1.5

$\text{rank}(A'A) \leq \text{rank}(A) = \text{rank}(P_A A)$ $P_A A = A$

(lem 2.2)

$= \text{rank}(A(A'A)^{-}A'A) \leq \text{rank}(A'A)$

(Lemma 1.5) (Lem 2.2)

What happens rank(X) = p* < p? what is estimable?

**Question.** How many essentially different estimable functions are there?

To answer the question, we make precise the notion of "essentially different."

**Definition.** Linear functions $\underline{\lambda}_1'\underline{\beta}, \ldots, \underline{\lambda}_r'\underline{\beta}$ are said to be linearly independent if $\underline{\lambda}_1, \ldots, \underline{\lambda}_r$ form a linearly independent set of vectors or, equivalently, if the matrix $\Lambda = (\underline{\lambda}_1, \ldots, \underline{\lambda}_r)$ has rank $r$.

$p^* \le$ # of col of $X = p =$ # of parameters $\beta$

rank(X)

**Theorem 2.2.** *Under Models II—VII, there exists a set of $p^*$ linearly independent functions $\underline{\lambda}_1'\underline{\beta}, \ldots, \underline{\lambda}_{p^*}'\underline{\beta}$ that are estimable. No collection of estimable linear functions can contain more than $p^*$ linearly independent functions.*

*Proof.* Let $\underline{u}_i'$ be the $i$th row of $X$. There exist $p^*$ linearly independent rows of $X$, say $\underline{u}_{k_1}', \ldots, \underline{u}_{k_{p^*}}'$. By Corollary 1 to Theorem 2.1, $\underline{u}_{k_1}'\underline{\beta}, \ldots, \underline{u}_{k_{p^*}}'\underline{\beta}$ are $p^*$ linearly independent estimable functions, proving the first part of the theorem.

Next, suppose $\underline{\lambda}_1'\underline{\beta}, \ldots, \underline{\lambda}_q'\underline{\beta}$ is some collection of estimable functions.

Stack these into a $q \times p$ matrix $\Lambda' = \begin{pmatrix} \underline{\lambda}_1' \\ \vdots \\ \underline{\lambda}_q' \end{pmatrix}$

where $\Lambda' = A'X$ must hold for some $q \times n$ matrix $A'$

Now $\text{rank}(\Lambda) = \text{rank}(\Lambda')$ ← Lemma 2.2

$= \text{rank}(A'X) \le \text{rank}(X) = p^*$

that is, # of lin independent columns of $\Lambda = (\underline{\lambda}_1, \ldots, \underline{\lambda}_q)$ can't be more than $p^*$

## 2.3 Least Squares Estimation

**Definition.** Under Models II—VII, a (simple or ordinary) least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is $\underline{\lambda}'\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any value of $\underline{\beta}$ that minimizes the sum of squares

$$Q(\underline{\beta}) = (\underline{y} - X\underline{\beta})'(\underline{y} - X\underline{\beta}).$$

A necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$ is that

$$\left.\frac{\partial Q(\underline{\beta})}{\partial \underline{\beta}}\right|_{\underline{\beta}=\widehat{\underline{\beta}}} = \underline{0}.$$

Using vector calculus (see, for example, Section 10.8 of Graybill's book *Matrices with Applications in Statistics*),

$$\begin{aligned}\frac{\partial Q(\underline{\beta})}{\partial \underline{\beta}} &= \frac{\partial(\underline{y}'\underline{y})}{\partial \underline{\beta}} - \frac{\partial(2\underline{y}'X\underline{\beta})}{\partial \underline{\beta}} + \frac{\partial(\underline{\beta}'X'X\underline{\beta})}{\partial \underline{\beta}} \\ &= -2X'\underline{y} + 2X'X\underline{\beta}.\end{aligned}$$

Thus, a necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$ is that $\widehat{\underline{\beta}}$ satisfy the equations

$$X'X\widehat{\underline{\beta}} = X'\underline{y}.$$

These equations are called the normal equations.

**Remark.** (The material in this remark is optional.) The vector-calculus results used above can be obtained as follows. For $\underline{\beta} = (\beta_1, \ldots, \beta_p)'$ and any $p \times 1$ vector of constants $\underline{a} = (a_1, \ldots, a_p)'$,

$$\frac{\partial \underline{a}'\underline{\beta}}{\partial \underline{\beta}} = \begin{pmatrix} \frac{\partial(a_1\beta_1 + \cdots + a_p\beta_p)}{\partial \beta_1} \\ \vdots \\ \frac{\partial(a_1\beta_1 + \cdots + a_p\beta_p)}{\partial \beta_p} \end{pmatrix} = \begin{pmatrix} a_1 \\ \vdots \\ a_p \end{pmatrix} = \underline{a}.$$

Thus,

$$\frac{\partial(\underline{y}'X\underline{\beta})}{\partial \underline{\beta}} = \frac{\partial\left[(X'\underline{y})'\underline{\beta}\right]}{\partial \underline{\beta}} = X'\underline{y}.$$

Let $\underline{u}_i'$ be the $i$th row of $X$. Then,

$$X\underline{\beta} = \begin{pmatrix} \underline{u}_1' \\ \vdots \\ \underline{u}_n' \end{pmatrix} \underline{\beta} = \begin{pmatrix} \underline{u}_1'\underline{\beta} \\ \vdots \\ \underline{u}_n'\underline{\beta} \end{pmatrix}$$

and

$$\underline{\beta}' X' X \underline{\beta} = (\underline{u}_1'\underline{\beta}, \ldots, \underline{u}_n'\underline{\beta}) \begin{pmatrix} \underline{u}_1'\underline{\beta} \\ \vdots \\ \underline{u}_n'\underline{\beta} \end{pmatrix} = \sum_{i=1}^{n} (\underline{u}_i'\underline{\beta})^2.$$

Thus,

$$\frac{\partial(\underline{\beta}' X' X \underline{\beta})}{\partial \underline{\beta}} = \frac{\partial}{\partial \underline{\beta}} \left( \sum_{i=1}^{n} (\underline{u}_i'\underline{\beta})^2 \right) = \sum_{i=1}^{n} 2(\underline{u}_i'\underline{\beta}) \frac{\partial(\underline{u}_i'\underline{\beta})}{\partial \underline{\beta}} = 2 \sum_{i=1}^{n} (\underline{u}_i'\underline{\beta}) \underline{u}_i$$

$$= 2 \sum_{i=1}^{n} \underline{u}_i (\underline{u}_i'\underline{\beta}) = 2 \sum_{i=1}^{n} (\underline{u}_i \underline{u}_i') \underline{\beta} = 2X' X \underline{\beta}.$$

**Theorem 2.3.** *The normal equations are consistent.* Can find a solution $\underline{b}$ to $X'X\underline{b} = X'\underline{y}$

Note: if NEs have a solution, then one solution must be $(X'X)^- X'\underline{y}$ too.

Actually, we can solve $X'XB = X'$ using $B = (X'X)^- X'$ Lemma 1.4

So, $\underline{b} = B\underline{y}$ solves

$X'X\underline{b} = \underbrace{X'XB}_{X'}\underline{y} = X'\underline{y}$ Note: $\underline{b} = (X'X)^- X'\underline{y}$

**Theorem 2.4.** *$Q(\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$ if and only if $\widehat{\underline{\beta}}$ is a solution to the normal equations.*

*Proof.* For the "if" part ($\Longleftarrow$), let $\widehat{\underline{\beta}}$ represent any solution to the normal equations. Then,

$$Q(\underline{\beta}) = (\underline{y} - X\underline{\beta})'(\underline{y} - X\underline{\beta}) = [(\underline{y} - X\widehat{\underline{\beta}}) + (X\widehat{\underline{\beta}} - X\underline{\beta})]'[(\underline{y} - X\widehat{\underline{\beta}}) + (X\widehat{\underline{\beta}} - X\underline{\beta})]$$

$$= [(\underline{y} - X\widehat{\underline{\beta}}) + X(\widehat{\underline{\beta}} - \underline{\beta})]'[(\underline{y} - X\widehat{\underline{\beta}}) + X(\widehat{\underline{\beta}} - \underline{\beta})]$$

$$= (\underline{y} - X\widehat{\underline{\beta}})'(\underline{y} - X\widehat{\underline{\beta}}) + [X(\widehat{\underline{\beta}} - \underline{\beta})]'[X(\widehat{\underline{\beta}} - \underline{\beta})] + 2(\widehat{\underline{\beta}} - \underline{\beta})'X'(\underline{y} - X\widehat{\underline{\beta}}).$$

For the "only if" part ($\Longrightarrow$), note that $Q(\underline{\beta}) \geq Q(\widehat{\underline{\beta}})$ for all $\underline{\beta}$ with equality holding only if $X(\widehat{\underline{\beta}} - \underline{\beta}) = \underline{0}$, that is, only if $X\underline{\beta} = X\widehat{\underline{\beta}}$. Thus, if $\widetilde{\underline{\beta}}$ is any other value of $\underline{\beta}$ at which $Q(\underline{\beta})$ attains a minimum, then $X\widetilde{\underline{\beta}} = X\widehat{\underline{\beta}}$.

From Theorem 2.4, $t(\underline{y})$ is a **least squares estimate** of an estimable function $\underline{\lambda}'\underline{\beta}$ if and only if $t(\underline{y}) = \underline{\lambda}'\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any solution to the normal equations.

For an estimable function $\underline{\lambda}'\underline{\beta}$, there remains the question of whether $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ to the normal equations, that is, whether $\underline{\lambda}'\underline{\beta}$ has a unique least squares estimate, and, more generally, whether for any function $\underline{\lambda}'\underline{\beta}$, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$. The answer is given by the following theorem.

**Theorem 2.5.** *Under Models II—VII, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ to the normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.*

*Proof.* For the "only if" part ($\Longrightarrow$), by Theorem 1.3, all solutions to the normal equations are obtained from

$$(X'X)^{-}X'\underline{y} + [I - (X'X)^{-}X'X]\underline{z}$$

as $\underline{z}$ ranges over the space of all possible vectors. Thus, setting $\widehat{\underline{\beta}} \equiv (X'X)^{-}X'\underline{y}$ as one solution and $\widetilde{\underline{\beta}} \equiv (X'X)^{-}X'\underline{y} + [I - (X'X)^{-}X'X]\underline{z}$ as another solution for some $\underline{z}$, if $\lambda'\widehat{\underline{\beta}} = \lambda'\widetilde{\underline{\beta}}$ holds for any two solutions of the normal equations, then

$$\underline{\lambda}'[I - (X'X)^{-}X'X]\underline{z} = 0 \text{ for all } \underline{z}.$$

This implies that

$$\underline{\lambda}'[I - (X'X)^{-}X'X] = \underline{0}' \text{ or } \underline{\lambda}' = \underline{\lambda}'(X'X)^{-}X'X = \underline{a}'X \text{ for } \underline{a} = [\underline{\lambda}'(X'X)^{-}X']'.$$

By Theorem 2.1, $\underline{\lambda}'\underline{\beta}$ is estimable.

**Corollary.** $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ to the normal equations.